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Singularities

Problem 37 Which of the following functions have a removable singularity at the given point z_0?

(a) $\dfrac{6-z-z^2}{2-z}$, $z_0 = 2$.

(b) $\dfrac{\sin z}{z^2-\pi^2}$, $z_0 = \pi$.

(c) $\dfrac{\sin z}{(z-\pi)^2}$, $z_0 = \pi$.

(d) $\dfrac{\sin z}{z^2+z}$, $z_0 = 0$.

Problem 38 Which of the following functions have a pole at the given point z_0? If z_0 is a pole, determine its order.

(a) $\dfrac{e^{(z^2)}-1}{z^3}$, $z_0 = 0$.

(b) $\dfrac{z}{1-\sin z}$, $z_0 = 0$.

(c) $\dfrac{\cos z - 1}{z^2}$, $z_0 = 0$.

(d) $\tan^2 z$, $z_0 = \frac{\pi}{2}$.

Problem 39 Recall that we say that a function F has a zero of order $n \in \mathbb{N}$ at a point z_0 if $F(z_0) = \cdots = F^{(n-1)}(z_0) = 0$, $F^{(n)}(z_0) \neq 0$.

Assume that $h(z) = \frac{f(z)}{g(z)}$, where the functions f and g are holomorphic in $D(z_0; R)$ with some $R > 0$, f has a zero of order $n \in \mathbb{N}$ at z_0, and g has a zero of order $m \in \mathbb{N}$ at z_0. Prove the following statements.

(a) If $n \geq m$, then z_0 is a removable singularity of h. Moreover,

$$\lim_{z \to z_0} h(z) = \frac{f^{(m)}(z_0)}{g^{(m)}(z_0)}.$$

(b) If $n < m$, then z_0 is a pole of h of order $m - n$.

Residues

Problem 40 Calculate residues of given functions at all their singularities.

(a) $\dfrac{2z^2+1}{z^2+25}$,

(b) $\dfrac{1}{\sin(\pi z)}$,

(c) $\dfrac{1-\cos z}{z^3}$,

(d) $\dfrac{z^2}{(1+z)^3}$.